

**PRETRAINING • 1,226 unlabeled UChicago exams**

**Vessel curves +  
elapsed times**



**Graph encoder**



**Forecasting  
decoder**



Future node  
enhancement

transfer weights

**FINE-TUNING • 179 exams / 141 patients**



**Same pretrained  
graph encoder**



**pCR prediction  
head**

compared with the identical  
randomly initialized encoder